

```

remnux@remnux:~$ NoMoreXor.py -a -o malware.hex malware.exe
[*] Attempting auto analysis
[*] HEXING.....: malware.exe
[*] Saving AS.....: malware.hex
[*] Attempting to guess the XOR key of.....: malware.hex
[*] Size of content.....: 27076210
[*] Total pages (3024):.....: 26441
[*] Total contiguous 512 chunks.....: 52852
[*] Top (5) overall chars
Occurrences | Character(s)
-----|-----
88441 = 0x00
71980 = 0xFF
61607 = 0xFE
61067 = 0xDF
60967 = 0xAF
[*] Total number of unique 512 chunks.....: 52077
[*] Top (5) 512 char sequences after cleanup

```

Figure 1: NoMoreXor

```

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Figure 2: NoMoreXor for extracting possible XOR keys in the sample 'malware.exe'

Balbuzzard

Balbuzzard is a malware analysis tool that is used to extract patterns from malicious files and to crack obfuscated code using XOR. It is also used to extract strings/patterns and embedded files.

The syntax is:

`Balbuzzard.py [option] [filename.extension]`

Here's an example:

`Balbuzzard.py -c -vsw malware.csv malware.exe`

Bbcrack

Bbcrack is a supplementary tool used to brute-force XOR keys based on patterns of interest.

The syntax is:

`bbcrack.py [option] [filename.extension]`

An example is:

`bbcrack.py -l 2 malware.exe`

Bbharvest

Bbharvest is also a brute force tool used to extract XOR keys that are specifically targeted against single

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remnux@remnux:~$ balbuzzard.py -csw malware.csv malware.exe
Loading plugin from bbz_sample.plugin.py
Loading yara plugin from clamsrc/clamsrc.yara
Loading yara plugin from malwarecookbook/capabilities.yara
Loading yara plugin from malwarecookbook/magic.yara
Loading yara plugin from Alienvaultlabs/urauy.skypedat.yara
Loading yara plugin from Alienvaultlabs/mask.yara
Loading yara plugin from Alienvaultlabs/leverage.yara
Loading yara plugin from Alienvaultlabs/hangover.yara
Loading yara plugin from Alienvaultlabs/APT_NGO_uuacit_PDF.yara
Loading yara plugin from Alienvaultlabs/APT_NGO_uuacit.yara
Loading yara plugin from Alienvaultlabs/GeorBotBinary.yara
Loading yara plugin from Alienvaultlabs/xint.yara
Loading yara plugin from Alienvaultlabs/apti.yara
Writing output to CSV file: malware.csv
Opening file malware.exe

```

Figure 3: Balbuzzard

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remnux@remnux:~$ bbbcrack.py -l 2 malware.exe
Loading transform plugin from /opt/remnux-balbuzzard/plugins/trans_
sample_plugin.py
Opening file malware.exe
STAGE 1: quickly counting simple patterns for all transforms
Best score so far: Identity, stage 1 score=27875360
Transform score for: stage 1 score=187776

```

Figure 4: Bbcrack

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remnux@remnux:~$ bbbharvest.py -t list malware.exe
Loading transform plugin from /opt/remnux-balbuzzard/plugi
ns/trans_sample.plugin.py
Available transforms - Level 1:
- Identity: Identity transformation, no change to data. P
arameters: none.
- xor: XOR with 8 bits static key A. Parameters: A (1-FF)
- add: ADD with 8 bits static key A. Parameters: A (1-FF)
- rol: ROL - rotate A bits left. Parameters: A (1-7).
- xor_rol: XOR with static 8 bits key A, then rotate 8 bi
ts left. Parameters: A (1-FF), 8 (1-7).
- add_rol: ADD with static 8 bits key A, then rotate 8 bi
ts left. Parameters: A (1-FF), 8 (1-7).
- rol_add: rotate A bits left, then ADD with static 8 bi
t key B. Parameters: A (1-7), 8 (1-FF).
Level 2:

```

Figure 5: Bbharvest

obfuscated strings/patterns in malware.

The syntax is:

`Bbharvest [option] [filename.extension]`

Here's an example:

`bbharvest.py -l 2 malware.exe`

Bbtrans

Bbtrans is used to apply transforms over the malware samples.

The syntax is:

`Bbtrans.py [option] [filename.extension]`

And an example is:

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